Консултация по БД – 01.06.2024

B1. Разпознаване на нормалните форми и процесът на преминаване от една нормална форма в друга

Нека е дадена релацията Company (deptno, deptname) и

1) FD-1: deptno->deptname

2) $\{deptno\}^+ = \{deptno, deptname\}$
$\{deptname\}^+ = \{deptname\}$
$\{deptno, deptname\}^+ = \{deptno, deptname\}$

deptno -> deptno, deptname ; deptno -> deptno ; deptno -> deptname ;
deptname -> deptname
deptno, deptname -> deptno ; deptno, deptname -> deptname;
deptno,deptname -> deptno, deptname;

3) Релацията Company е в НФБК

Нека е дадена релацията R (A, B, C) и

FD-1: A -> B
FD-2: B -> C

$\{\}^+ = \{\}$
$\{A\}^+ = \{A, B, C\}$ – key
$\{B\}^+ = \{B, C\}$
$\{C\}^+ = \{C\}$
$\{A, B\}^+ = \{A, B, C\}$ – super key
$\{A, C\}^+ = \{A, C, B\}$ – super key
$\{B, C\}^+ = \{B, C\}$
$\{A, B, C\}^+ = \{A, B, C\}$ – super key

B -> B ; B -> C; B -> BC; - нарушават 3НФ => релацията R е в 2НФ
C -> C;
B, C -> B;
B, C -> C;
B, C -> B, C;

Декомпозираме R (A, B, C) → R1(B, C); R2(B, A); R1 и R2 са в НФБК.

В2. Понеже в анотацията на тема 23, за релационния модел на данните, в частта за релационната алгебра се говори за оператор частно, който не сме споменавали на лекции, може ли да го изпуснем без това да повлияе отрицателно?

Нека е дадена R (A, B) и S(B) – атрибутите на S трябва да са подмножество на тези на R; Частно R÷S е релация, която се състои:

1) само от тези атрибути на R, които не са в S
2) и от тези кортежи на R, проектирани по атрибутите от 1), такива че при декартовото им произведение с кортежите от релацията S, резултатния кортеж да принадлежи на R

R

A	B
a	1
b	2
a	2

S

B
1
2

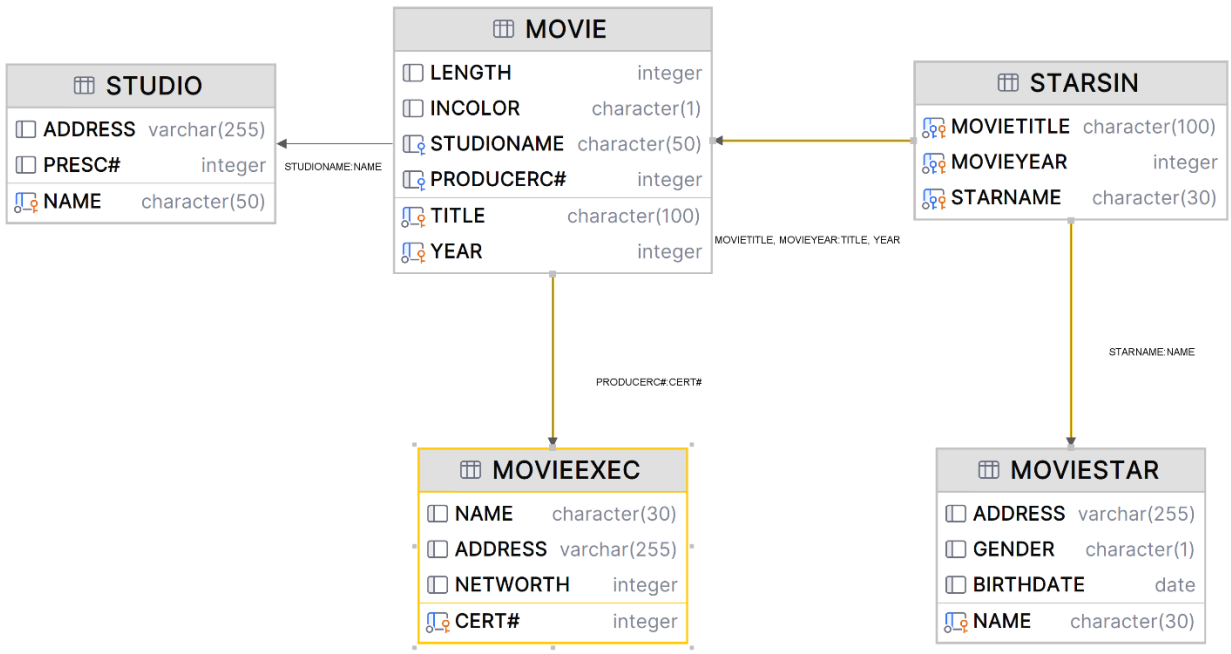
R÷S

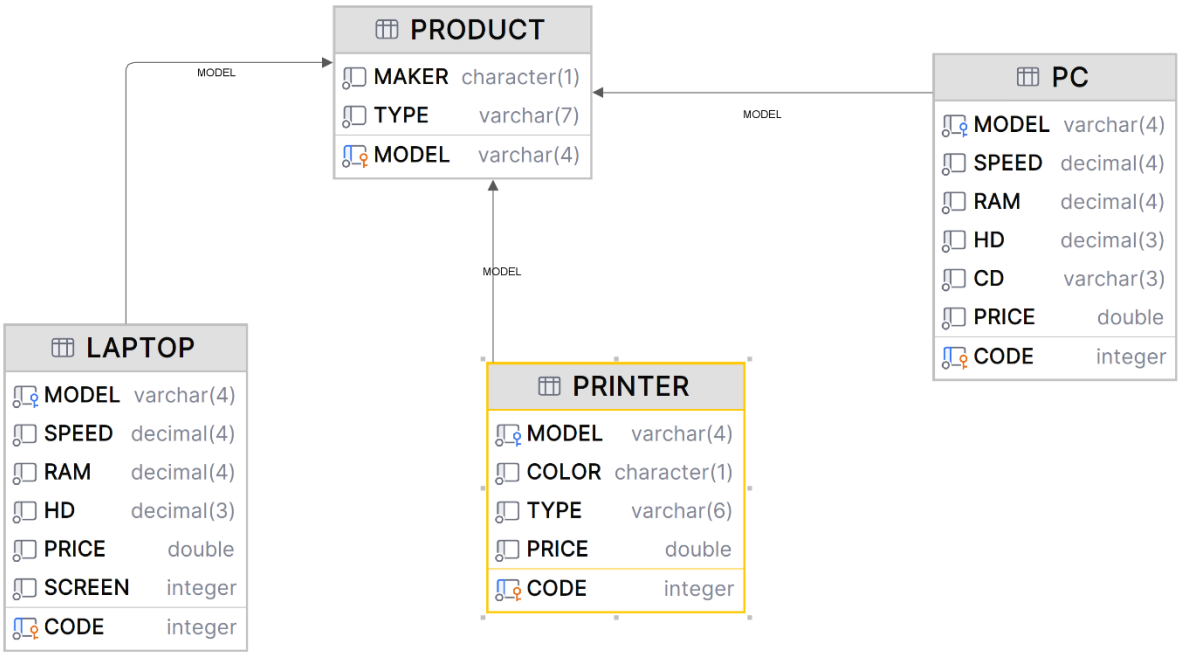
A
a

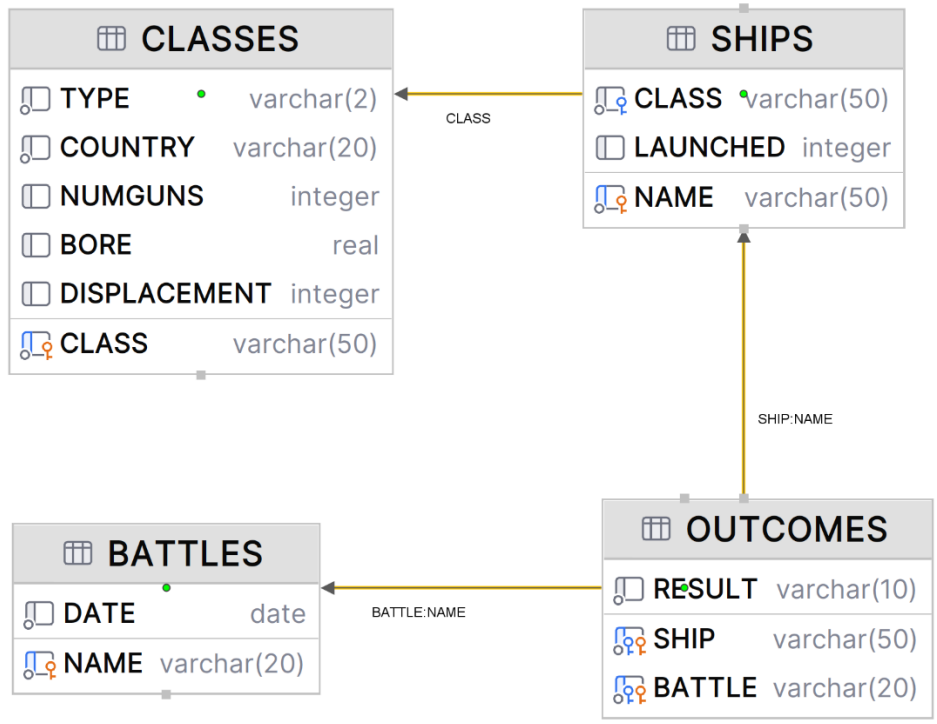
Книга - Database Systems - The Complete Book (2nd Edition):
https://people.inf.elte.hu/kiss/DB/ullman_the_complete_book.pdf

Tool Functional Dependencies Checker: https://arjo129.github.io/functionalDependencyCalculator/

Схеми на БД




PRODUCT
MAKER character(1)
TYPE varchar(7)
MODEL varchar(4)
PC
MODEL varchar(4)
SPEED decimal(4)
RAM decimal(4)
HD decimal(3)
CD varchar(3)
PRICE double
CODE integer
LAPTOP
MODEL varchar(4)
SPEED decimal(4)
RAM decimal(4)
HD decimal(3)
PRICE double
SCREEN integer
CODE integer
PRINTER
MODEL varchar(4)
COLOR character(1)
TYPE varchar(6)
PRICE double
CODE integer
MODEL
MODEL
MODEL


CLASSES
TYPE varchar(2)
COUNTRY varchar(20)
NUMGUNS integer
BORE real
DISPLACEMENT integer
CLASS varchar(50)
SHIPS
CLASS varchar(50)
LAUNCHED integer
NAME varchar(50)
CLASS
SHIP:NAME
BATTLES
DATE date
NAME varchar(20)
OUTCOMES
RESULT varchar(10)
SHIP varchar(50)
BATTLE varchar(20)
BATTLE:NAME

Решения на задачи от практическата част на ДИ

```sql
-- 11.09.2023

-- 1. year <= 2000 and year > 2000

SELECT STARNAME
FROM STARSIN S, MOVIESTAR MS
WHERE GENDER = 'F' AND S.STARNAME = MS.NAME
AND MOVIEYEAR <= 2000
INTERSECT
SELECT STARNAME
FROM STARSIN S, MOVIESTAR MS
WHERE GENDER = 'F' AND S.STARNAME = MS.NAME
AND MOVIEYEAR > 2000;


-- II

SELECT DISTINCT STARNAME
FROM STARSIN S, MOVIESTAR MS
WHERE GENDER = 'F' AND S.STARNAME = MS.NAME
AND MOVIEYEAR <= 2000 AND MS.NAME IN (SELECT STARNAME
                                      FROM STARSIN S, MOVIESTAR MS
                                      WHERE GENDER = 'F'
                                      AND S.STARNAME = MS.NAME
                                      AND MOVIEYEAR > 2000);

-- III

SELECT MS.NAME
FROM STARSIN S, MOVIESTAR MS
WHERE GENDER = 'F' AND S.STARNAME = MS.NAME
GROUP BY MS.NAME
HAVING MIN(S.MOVIEYEAR) <= 2000 AND MAX(S.MOVIEYEAR) > 2000;

-- 11.07.2023

SET SCHEMA DB2SHIPS;

-- I

SELECT C.CLASS
FROM CLASSES C LEFT JOIN SHIPS S ON C.CLASS = S.CLASS
WHERE S.NAME IS NULL
UNION
SELECT S.CLASS
FROM SHIPS S LEFT JOIN OUTCOMES O ON S.NAME = O.SHIP
WHERE O.SHIP IS NULL
EXCEPT
SELECT CLASS FROM SHIPS, OUTCOMES WHERE NAME = SHIP AND RESULT = 'damaged';

-- II

SELECT CLASS
FROM CLASSES
WHERE CLASS NOT IN (SELECT CLASS FROM SHIPS JOIN OUTCOMES ON NAME = SHIP
WHERE RESULT = 'damaged');
```

```sql
-- 09.2022

SET SCHEMA DB2MOVIE;

-- TITLE, LENGTH, INCOLOR = 'Y'

SELECT TITLE, LENGTH, INCOLOR
FROM MOVIE
WHERE INCOLOR = 'Y'
AND (LENGTH NOT IN (SELECT MAX(LENGTH) FROM MOVIE WHERE INCOLOR = 'Y')
AND LENGTH IS NOT NULL);


-- 09.2020

-- 1
SELECT NAME, ADDRESS FROM STUDIO, MOVIE
WHERE NAME = STUDIONAME AND INCOLOR = 'Y'
INTERSECT
SELECT NAME, ADDRESS FROM STUDIO, MOVIE
WHERE NAME = STUDIONAME AND INCOLOR = 'N'
ORDER BY ADDRESS;

-- 2
SELECT NAME, ADDRESS, AVG(LENGTH) AS AVG_LEN, COUNT(TITLE) AS CNT
FROM STUDIO S LEFT JOIN MOVIE M ON S.NAME = M.STUDIONAME
GROUP BY NAME, ADDRESS
HAVING COUNT(TITLE) <= 3;
```